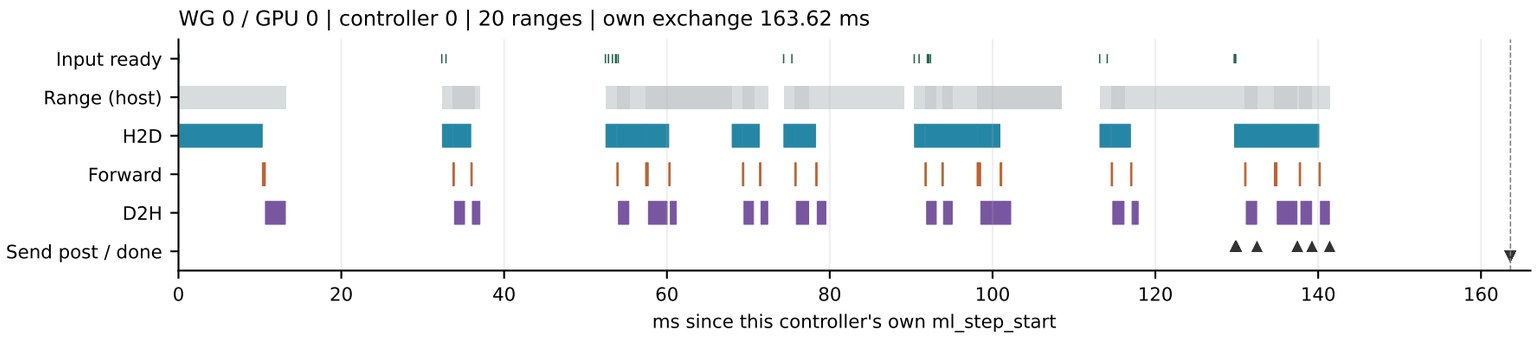
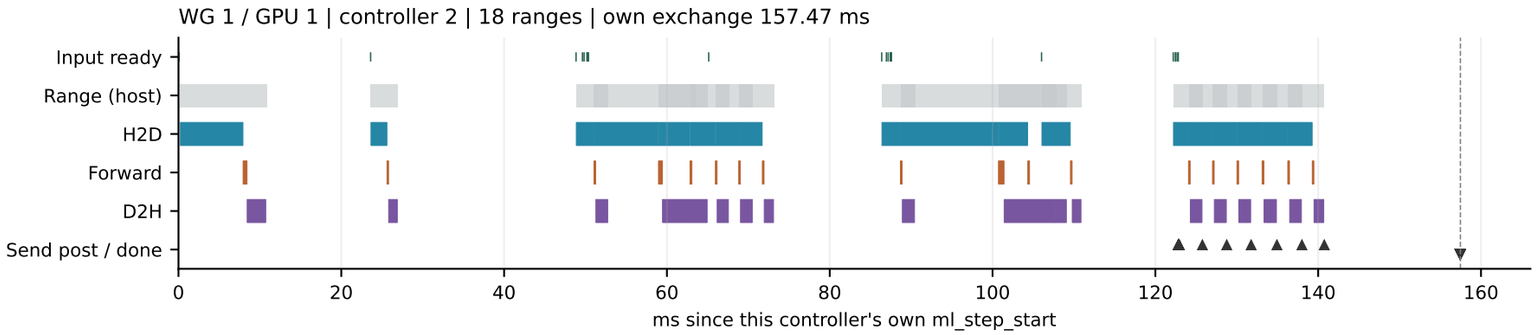
mlp | 96c4g | numa | collective vs p2pfull | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 388.14 ms | p2pfull 332.69 ms



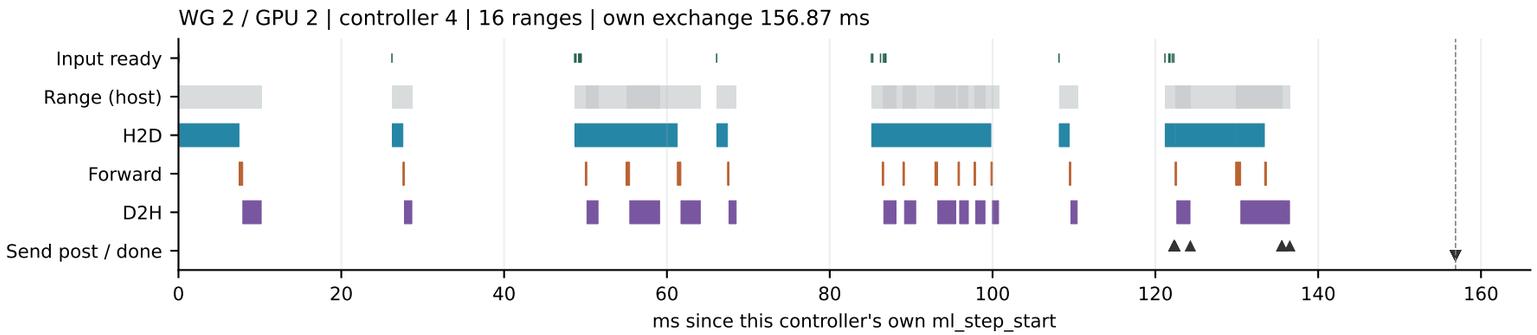
GPU work (ms)	Collective	p2pfull
H2D	27.846	52.692
Forward	2.890	4.482
D2H	15.810	27.818
GPU SUM	46.546	84.992

Controllers: coll 0 / P2P 0
Coll host device elapsed: 46.745 ms
P2P GPU envelope: 141.220 ms
Different scopes; not an elapsed-time comparison.



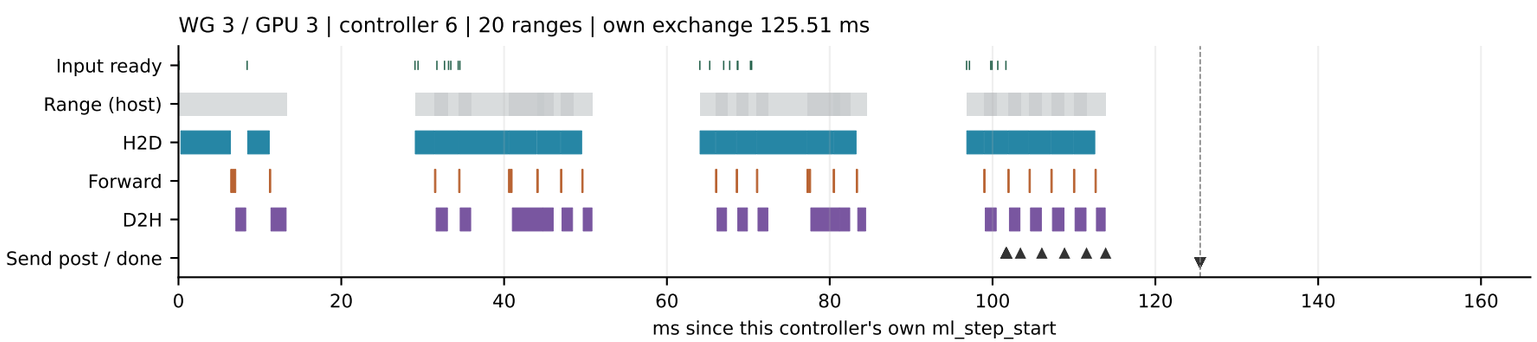
GPU work (ms)	Collective	p2pfull
H2D	29.922	70.163
Forward	2.904	4.758
D2H	20.263	32.975
GPU SUM	53.090	107.896

Controllers: coll 2 / P2P 2
Coll host device elapsed: 53.355 ms
P2P GPU envelope: 140.445 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	29.335	50.165
Forward	2.896	4.371
D2H	14.844	27.919
GPU SUM	47.075	82.455

Controllers: coll 4 / P2P 4
Coll host device elapsed: 47.282 ms
P2P GPU envelope: 136.352 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	32.147	63.166
Forward	2.724	4.559
D2H	13.972	30.314
GPU SUM	48.843	98.040

Controllers: coll 6 / P2P 6
Coll host device elapsed: 49.041 ms
P2P GPU envelope: 113.547 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.